39. Let "$\cdot$" represent the scalar product and "$\times$" the vector product. Which of the following operations gives the magnitude of the projection of vector $\vec{a}$ on vector $\vec{b}$?

A. $\vec{a} \times \frac{\vec{b}}{|\vec{b}|}$

B. $\vec{a} \times \vec{b}$

C. $\vec{a} \cdot \vec{b}$

D. $\vec{a} + \vec{b}$

E. $\vec{a} \cdot \frac{\vec{b}}{|\vec{b}|}$

40. A train travels at a speed of 144 km/h. Its wheels perfectly adhere to the rails (no sliding) and have a diameter $d = 80$ cm. How many turns do they do in one second, roughly?

A. 8

B. 57

C. 32

D. 115

E. 16